

Week 7 RAG Conversation Summary

Summary of the technical discussion held during implementation.

pgvector

Enabled the PostgreSQL vector extension and prepared the database for vector embeddings.

Google Gen AI SDK

Discussed the difference between the SDK and Gemini models. The SDK communicates with Google's services; the models perform chat or embedding generation.

Embedding Generation

Implemented GeminiEmbedder. Learned that each call generates a new embedding for the supplied text or chunk. The response contains a list because batch embedding is also supported.

Chunking

Implemented DocumentChunker with RecursiveCharacterTextSplitter, explored chunk_size, chunk_overlap, enumerate(), Chunk dataclass, and metadata generation.

Overlap Discussion

Explained why chunk_overlap is a target rather than an exact guarantee and why text.find() gives more accurate character offsets than manually estimating positions.

DocumentChunk Model

Reviewed the SQLAlchemy model storing chunk text, embeddings, metadata, chunk order, timestamps, and the relationship to Document.

Testing & Debugging

Discussed test_chunker.py versus pytest, assert statements, Python slicing, and fixing a Uvicorn import-path error.